Burned-area mapping with Sentinel-2
dNBR/Otsu vs Random Forest

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1. Summary

I mapped the burned area of a 2022 wildfire from a pair of Sentinel-2 images (before and after the fire) and compared a classical, non-AI method (a threshold on the dNBR index) with an AI method (a Random Forest). Both were scored against MODIS burned-area polygons, on pixels that neither method had seen during training.

Main findings:

- The Random Forest beats the thresholds: pooled validation IoU of 0.792 against 0.725 for Otsu and 0.700 for the supervised threshold.
- After morphological clean-up applied to every method, the Random Forest reaches IoU 0.873 (F1 0.932), against 0.778 for Otsu and 0.799 for the tuned threshold.
- Most of the Random Forest's advantage comes from one extra input: a local spatial-context feature, not from model complexity.
- Overfitting was measured and reduced. A small train/validation gap remains in one of the two folds.

2. Question

Can the area burned by the 2022 fire be mapped automatically from Sentinel-2 images, and does a machine-learning classifier do it better than a simple spectral threshold?

The project is done alone, so one non-AI method is required and an AI method is optional. I implemented both, plus a stricter non-AI baseline, to make the comparison meaningful.

3. Data

3.1 What I used

Data	Source	Role
Sentinel-2 L2A, pre-fire, 2 July 2022 (tile T30TXQ)	Copernicus Data Space (ESA)	Input image
Sentinel-2 L2A, post-fire, 5 September 2022 (same tile)	Copernicus Data Space (ESA)	Input image
Scene Classification Layer (SCL) of both dates	Part of the Sentinel-2 L2A product	Cloud, shadow and no-data mask
Copernicus EMS observedEventA polygons	Copernicus Emergency Management Service	Used only to define the study area (ROI)
MODIS burned-area polygons (modis.ba.poly.shp)	NASA MODIS	**Target** (reference burned area)

Bands used: B02, B03, B04, B08 (visible and near infrared) at 10 m; B11 and B12 (shortwave infrared) at 20 m, resampled to the common 10 m grid.

3.2 Why these data

- **Sentinel-2 is free and sharp (10 m)**, and has the shortwave-infrared band B12. Burned vegetation loses its near-infrared reflectance and gains shortwave-infrared reflectance, so the pair B08/B12 is the standard way to see a burn scar.
- **A before/after pair** removes the part of the signal that is just land cover. What remains is the change caused by the fire.

- **MODIS burned-area polygons** give an independent reference that is not derived from my own images or my own method, which keeps the evaluation honest.
- **The Sentinel-2 scenes are one month apart and acquired under similar sun and season conditions**, so differences between them are mostly due to the fire.

3.3 What I would have chosen with a 1 M€ project

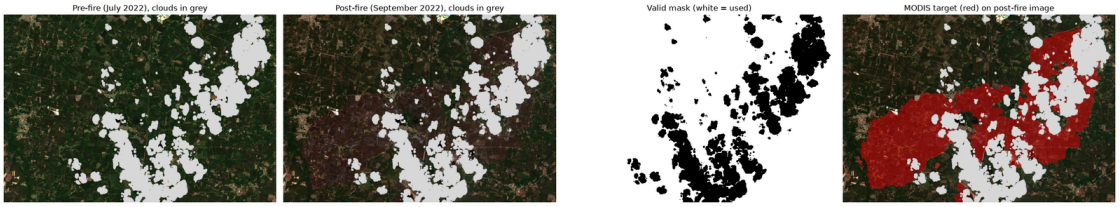
These are the limits of my current data and what I would change to remove them:

- **Better reference data.** MODIS burned-area maps come from roughly 500 m data, so their edges are coarse next to 10 m pixels. With a budget I would fund a **field campaign and drone or very-high-resolution imagery** (for example commercial 0.5 m optical images) to build a precise reference map over a sample of the burn.
- **Denser time series.** Combine Sentinel-2 with **Landsat and Sentinel-1 SAR**. SAR sees through clouds and smoke, and a dense time series would let me date the fire day by day and avoid the 18% of pixels lost to clouds.
- **Several fires and regions.** Train and test on many fires in different climates and vegetation types, so that the result is a general method and not a result for one event.
- **Burn severity labels** (for example from field surveys), to map how badly each area burned rather than only burned versus not burned.

4. Preparation of the data

1. **Study area (ROI).** The bounding box of the two Copernicus EMS polygons, plus a 1.5 km buffer. At 10 m resolution this gives an image of **3054 × 2101 pixels**.
2. **Reflectance conversion.** The products have processing baseline N0510 (4.00 or later), so reflectance is (DN − 1000) / 10000. This matters for ratio indices such as NBR, which are biased if the offset is ignored.
3. **Cloud mask.** A pixel is excluded if the SCL marks it as no-data, saturated, cloud shadow, medium or high-probability cloud, or cirrus in **either** date. The mask is then grown by 3 pixels (30 m) to remove cloud edges. This removes **18.42%** of the ROI.
4. **Target.** MODIS polygons are clipped to the ROI and rasterised on the 10 m grid, with the masked pixels set to "not used". The target contains **1,577,372 burned pixels**, after losing 548,195 under clouds.
5. **Masked pixels are ignored everywhere**: in the Otsu statistics, in training, in validation, in the metrics and in the figures (shown in grey).

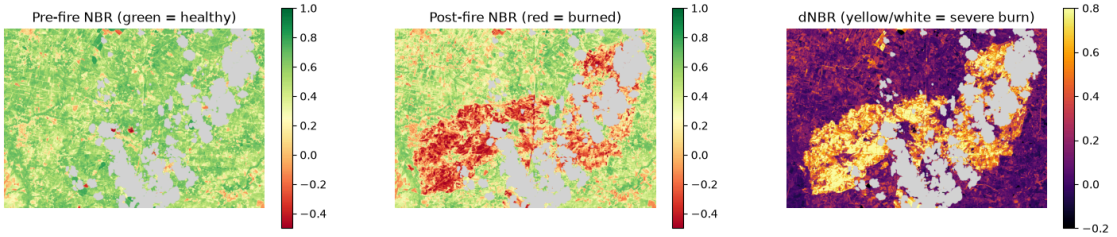
The image below contains Pre-fire (July 2022) and post-fire (September 2022) RGB images with clouds in grey, the valid mask (white = used), and the MODIS target (red) on the post-fire image.



5. Methods

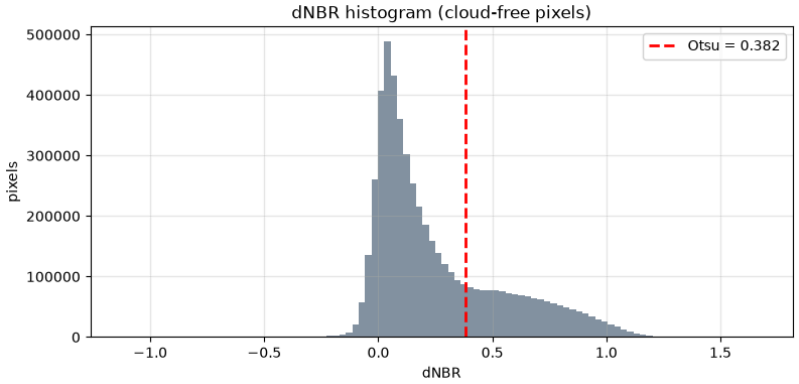
5.1 Spectral features

- **NBR** = (B08 − B12) / (B08 + B12), computed for the pre- and post-fire images.
- **dNBR** = NBR(pre) − NBR(post). A high value means vegetation was lost between the two dates, which is the burn signal.
- Candidate extra features, tested in section 7.3: NDVI (and its difference), the raw bands, and two **local-context features** (the mean and standard deviation of dNBR in an 11 × 11 pixel window, computed without using cloudy pixels).

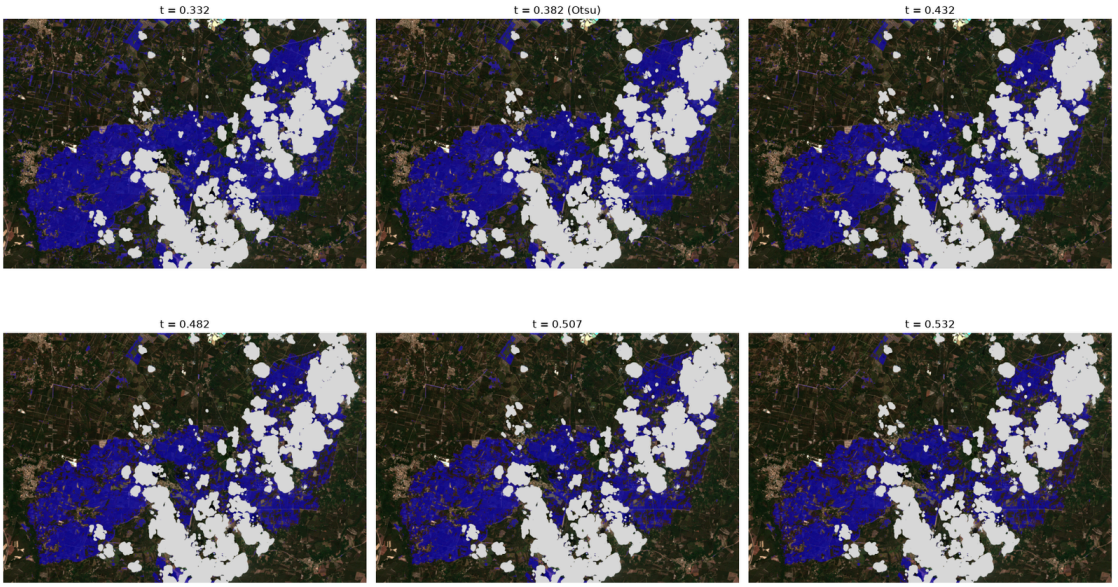


5.2 Non-AI method 1: Otsu threshold on dNBR

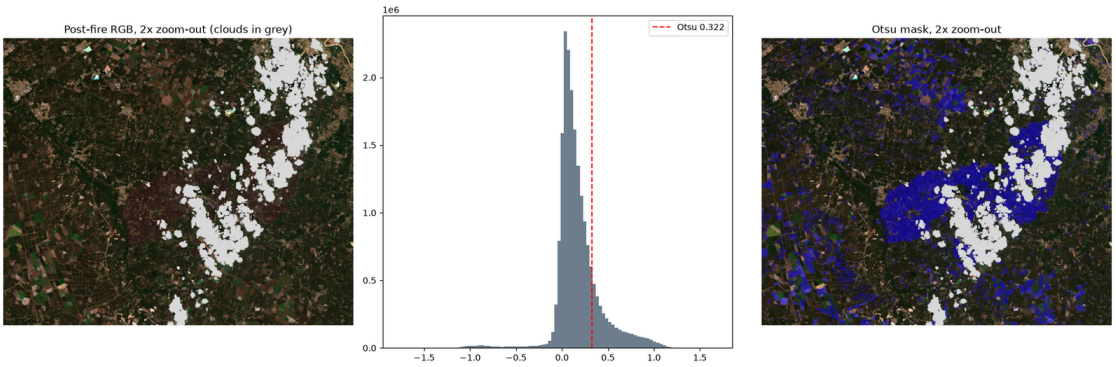
Otsu's method chooses automatically the threshold that best separates the dNBR histogram into two classes. It is **unsupervised**: it never sees the target. Pixels above the threshold are labelled burned. The threshold found is **0.382**. I also looked at how the map changes with the threshold, and repeated the whole pipeline on an area twice as large to check that the method still behaves sensibly.



Burned mask (blue) on the post-fire image for six thresholds around the Otsu value, showing how sensitive the map is to the threshold.



The same automatic pipeline on an area twice as large: post-fire RGB, dNBR histogram with its own Otsu threshold, and the resulting mask.



5.3 Non-AI method 2: supervised dNBR threshold

The same single-threshold idea, but the threshold is chosen to maximise IoU on the **training half** and then applied to the validation half. It is a stricter non-AI competitor than Otsu, and it separates the question "what does a threshold on dNBR give?" from "what does machine learning add?". The tuned thresholds were 0.382 (A) and 0.292 (B).

5.4 AI method: Random Forest

A Random Forest classifier predicts, for each pixel, the probability of being burned from its features. Final settings: **100 trees, maximum depth 6, minimum 20 samples per leaf**, trained on 300,000 randomly chosen training pixels per fold. These settings were chosen from train-versus-validation curves (section 7.2), not from the final score.

5.5 Morphological post-processing

Raw maps are speckled with isolated pixels and small holes. One identical clean-up is applied to **every** method so that the comparison stays fair:

1. delete isolated groups of fewer than 50 pixels;
2. opening with a disk of radius 5 (removes specks and thin bridges);
3. closing with a disk of radius 10 (merges nearby patches);
4. remove blobs smaller than 5,000 pixels
5. fill holes smaller than 5,000 pixels;
6. re-apply the cloud mask.

This step uses only the predicted mask and never the target, so it cannot leak labels.

6. Evaluation protocol

This is the part that makes the comparison trustworthy.

- **Two spatial folds.** Fold A trains on the left half of the image and validates on the right half. Fold B does the reverse. A random pixel split would be misleading, because neighbouring pixels are almost identical and the model would be tested on pixels next to ones it has seen.
- **Metrics are computed on validation pixels only**, for all methods, so they are compared on exactly the same pixels. Pooled means both validation halves together, so every pixel of the ROI is predicted by a model that never saw it.
- **Nothing is chosen on the data it is reported on.** The supervised threshold is tuned on the training half; the Random Forest settings come from the diagnostic curves.
- **Metrics.** IoU (overlap between prediction and target) is the main number, with F1, precision and recall. ROC-AUC and PR-AUC are also given for anything that produces a score. The burned class is a minority (27.5% of training pixels in fold A and 33.4% in fold B), so ROC-AUC looks flattering and **IoU and PR-AUC are the numbers to trust**.

7. Results

7.1 Main comparison (pooled validation pixels)

Method	IoU	F1	Precision	Recall	ROC-AUC	PR-AUC
Otsu	0.725	0.841	0.892	0.795	0.950	0.907
Otsu + morphology	0.778	0.875	0.968	0.799		
dNBR threshold (tuned on train)	0.700	0.824	0.817	0.830	0.950	0.907
dNBR threshold + morphology	0.799	0.888	0.944	0.839		
Random Forest	**0.792**	**0.884**	0.889	0.879	**0.972**	**0.947**
Random Forest + morphology	**0.873**	**0.932**	0.954	0.911		

IoU for each fold separately, to show how stable the result is:

Method	Fold A	Fold B	Pooled
Otsu	0.723	0.727	0.725
Otsu + morphology	0.722	0.830	0.778
dNBR threshold (tuned on train)	0.723	0.683	0.700
dNBR threshold + morphology	0.722	0.866	0.799
Random Forest	0.798	0.787	0.792
Random Forest + morphology	0.846	0.897	0.873

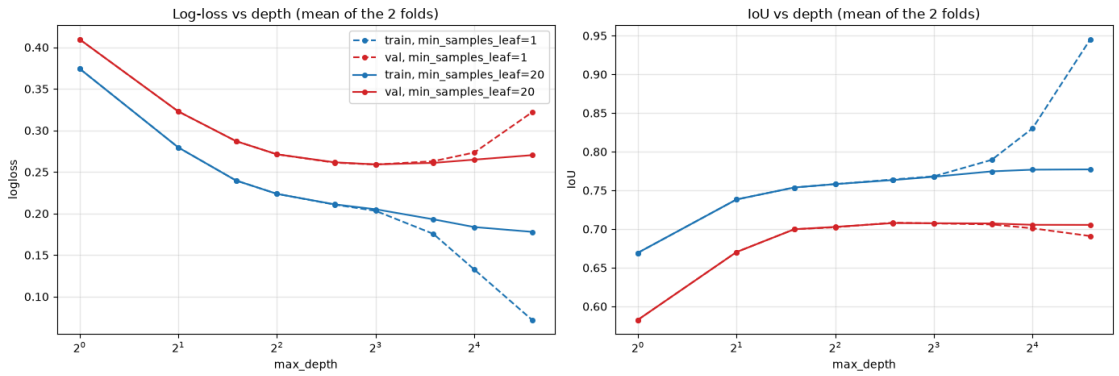
7.2 Is the Random Forest overfitting?

A forest has no epochs, so I drew train and validation log-loss against its two complexity settings, tree depth and number of trees. Validation loss was lowest at depth 8; the simplest depth within 2% of that minimum was **depth 6**, which I kept. Adding trees past a few tens did not widen the gap.

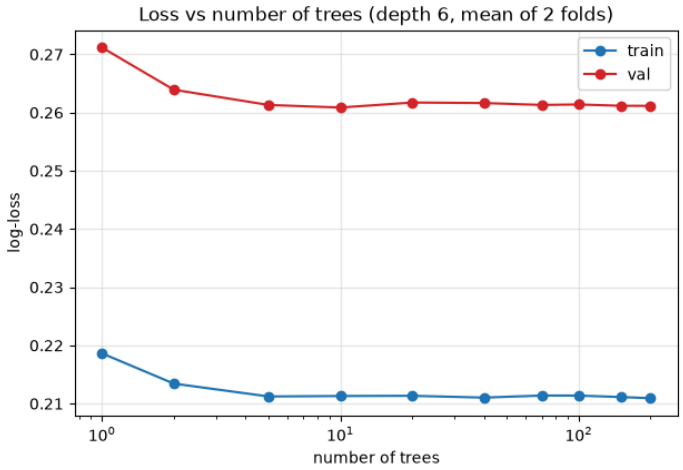
Configuration	Fold	IoU train	IoU validation	Gap
Original (depth 10, leaf 1)	A	0.766	0.703	0.063
Original (depth 10, leaf 1)	B	0.786	0.710	0.076
Regularised (depth 6, leaf 20)	A	0.754	0.703	0.051
Regularised (depth 6, leaf 20)	B	0.773	0.715	0.058

Regularising reduced the gap and cost nothing on validation, so the simpler forest is preferable. A gap above 0.05 is flagged as overfitting; these values are only just above that line.

Below is the Log-loss and IoU against tree depth, train (blue) and validation (red), for min_samples_leaf = 1 (dashed) and 20 (solid). Mean of the two folds.



Train and validation log-loss against the number of trees at depth 6. Mean of the two folds.



7.3 Are the input features good enough?

Same forest, same pixels, same two folds, only the input features change (mean of the two folds):

Feature set	Features	IoU train	IoU validation	PR-AUC validation	Gap
dNBR only	1	0.745	0.694	0.914	0.051
NBR pre, post and dNBR (original)	3	0.763	0.708	0.931	0.056
+ local context	5	0.846	**0.789**	**0.962**	0.056
+ NDVI	6	0.777	0.712	0.933	0.064
+ NDVI + raw bands	18	0.841	0.764	0.954	0.077

The rule for choosing was: keep the smallest set within 0.01 IoU of the best validation IoU. That selected **"+ local context"** (5 features).

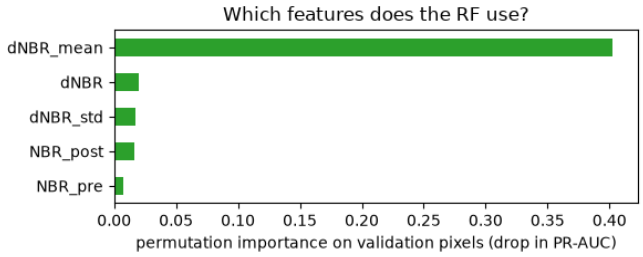
Below is the train and validation IoU for each feature set (mean of the two folds).



7.4 Final Random Forest, per fold

Fold	IoU train	IoU validation	Gap
A (train left, validate right)	0.834	0.798	0.036
B (train right, validate left)	0.857	0.787	0.070

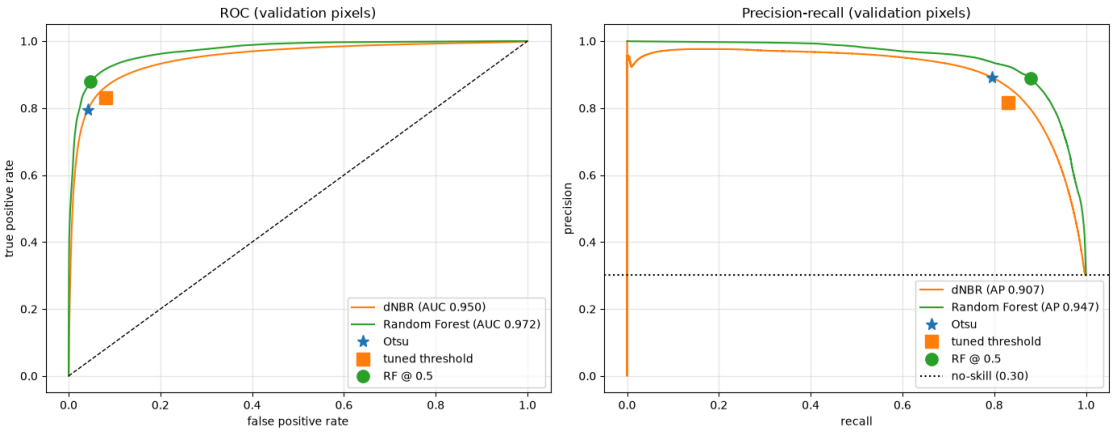
Permutation importance of each input feature of the final forest, measured on validation pixels (drop in PR-AUC).



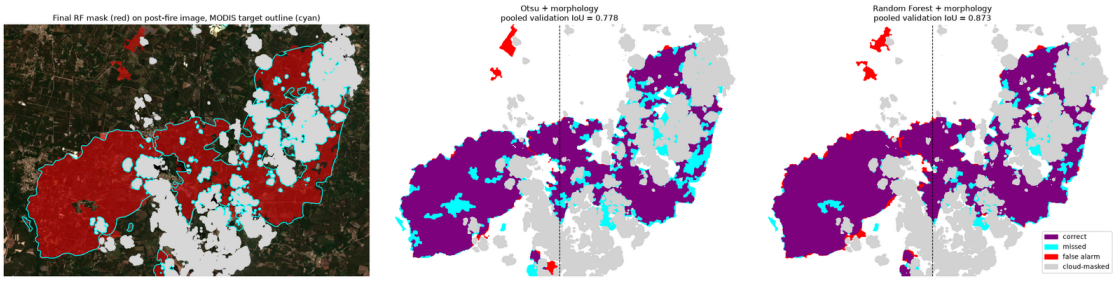
7.5 Visual results

Figures above show the data, the features, the threshold, and the diagnostics of the Random Forest. The two figures below summarise the final results: the ROC and precision-recall curves, and the maps of correct, missed and false-alarm pixels.

ROC curve (left) and precision-recall curve (right) on the pooled validation pixels. The markers show the operating point of Otsu, the tuned threshold and the Random Forest at 0.5



Left: final Random Forest mask (red) on the post-fire image with the MODIS target outline (cyan). Centre and right: error maps for Otsu and Random Forest, both after morphology. Purple = correct, cyan = missed, red = false alarm, grey = cloud-masked. The dashed line is the fold boundary.



8. Analysis

1. The AI method is better, but how much depends on the comparison. Before clean-up, the Random Forest gains about 0.07 IoU over Otsu and 0.09 over the tuned threshold. It improves both precision-recall quality (PR-AUC 0.947 against 0.907) and recall (0.879 against 0.795 for Otsu). The threshold-free scores show the advantage is real and not an artefact of where a threshold was placed.

2. The gain comes mostly from spatial context, not from a more complex model. On the same forest, going from dNBR only to the original three features gives 0.694 to 0.708, whereas adding the local mean and standard deviation of dNBR gives 0.789. A threshold looks at one pixel at a time, while a burn scar is a spatially coherent patch. The context features let the model use that. Adding NDVI or all raw bands did not help and mostly increased the overfitting gap, so more features is not better.

3. Morphology helps every method, and narrows the gap. The same clean-up reduces the number of separate blobs from thousands (7,049 for Otsu, 10,115 for the tuned threshold, 1,896 for the forest) to 12 to 15, and raises IoU by 0.05 to 0.10. Cleaned thresholds (0.778 and 0.799) come close to the *uncleaned* Random Forest (0.792), so a large part of what looks like "AI advantage" in raw maps is simply speckle. The Random Forest remains the best after clean-up (0.873).

4. The tuned threshold is not better than Otsu. The tuned threshold gave 0.700 against 0.725 for Otsu on the pooled data. The optimal threshold differs between the two folds (0.382 and 0.292), which means a single global dNBR threshold is not stable across this scene.

5. Overfitting is moderate and was checked. The final forest has a train-validation gap of 0.036 in fold A and 0.070 in fold B. Fold B is above my 0.05 warning level, so some overfitting remains. Validation IoU is nevertheless similar in both folds (0.798 and 0.787), so the conclusion does not depend on which half is used for training.

6. Morphology is not equally useful in both folds. In fold A it changes the thresholds' IoU by almost nothing (0.723 to 0.722) but raises the forest from 0.798 to 0.846. In fold B all methods gain substantially. The pooled gains should therefore be read as an average of two different situations.

9. Limitations

- **The target is coarse.** MODIS polygons come from about 500 m data, so their boundaries are much less precise than 10 m pixels. IoU against this target is capped for every method, and a method that scores lower may in fact be more accurate. Compact, blocky shapes (as produced by morphology) happen to resemble the target better, which partly explains the gains in section 7.1.
- **Clouds.** 18.42% of the area is ignored, including 548,195 burned target pixels. Results do not say anything about those pixels. The cloud-shadow class can catch dark burn scars; the notebook warns about this and it can be removed from the mask.
- **One fire, one pair of images.** The result may not carry over to other fires, seasons or vegetation types.
- **Spatial folds are left versus right halves.** Both halves contain the same kind of terrain, but a very different region would be a harder test. The local-context window (11 pixels) also crosses the fold boundary for a few pixels, but it uses image values only, never labels.
- **Morphology parameters were set by hand** and not tuned, which keeps them label-free but means they are not optimal.

10. Conclusion

A single threshold on dNBR is a fast, label-free, interpretable way to map the burn scar and already gives an IoU of about 0.73 against a coarse reference. A Random Forest that also sees the local spatial context of dNBR does clearly better (IoU 0.79 raw, 0.87 after clean-up), and I have shown with train-versus-validation tests that this gain is not caused by memorising the training pixels. The biggest practical improvements came from three simple choices rather than from model power: masking clouds everywhere, evaluating on spatially separate validation pixels, and using spatial context and post-processing. The main limit of the whole study is the quality of the reference data, which is where additional resources would be best spent.

11. Project journey: what I tried, what failed and what worked

The result in section 7 came after eleven steps, three of which failed or were dropped. The turning point was step 3: the first target, the Copernicus EMS polygons, turned out to be incomplete, so I changed the target and not only the method.

11.1 Step-by-step log

#	What I did	Result	What I learned	Next step I took
1	Downloaded Sentinel-2 L2A images (pre-fire 2 July 2022, post-fire 5 September 2022, tile T30TXQ) from the Copernicus Data Space, plus the Copernicus EMS observedEventA polygons. Cut a 3054 × 2101 pixel study area (EMS bounding box plus 1.5 km).	Worked	A good pair of dates, and the EMS polygons looked like a ready-made reference.	Compute dNBR.
2	Computed dNBR and applied an Otsu threshold on the whole area. Checked nearby thresholds by eye and repeated the pipeline on an area twice as large.	Worked	The burn scar is clearly visible, and Otsu is automatic and behaves sensibly on the larger area.	Measure the result against a reference.
3	Rasterised the EMS polygons as the target. Trained a Random Forest on one half of the image and validated on the other half.	Failed	Burned class: precision 0.61, recall 0.56, F1 0.59. The middle of the scene has no EMS polygons, so burned pixels there are scored as “unburned”. The target is incomplete.	Try to work around the gap.
4	Two-branch experiment: removed the empty middle strip (399 columns), split the rest into LEFT and RIGHT, and evaluated each side with a Random Forest trained on the other side.	Dropped	Mean IoU: Otsu 0.615, Random Forest 0.569. Scores differed a lot between sides (Otsu 0.656 and 0.574; Random Forest 0.652 and 0.486), so I could not separate method error from target error.	Stop patching the EMS target and look for a better one.
5	Replaced the target with the MODIS burned-area polygons clipped to the same area (7 polygons, 1,577,372 burned pixels). EMS is now used only to define the area. Added a cloud mask from the Scene Classification Layer, ignored everywhere (Otsu, training, validation, figures).	Worked	18.42% of the area is cloudy and 548,195 burned target pixels lie under clouds; they are now excluded. Otsu on the validation half: precision 0.94, recall 0.74, F1 0.83.	Compare Otsu and the Random Forest on cloud-free pixels.

#	What I did	Result	What I learned	Next step I took
6	Compared Otsu with the Random Forest (inputs: NBR pre, NBR post, dNBR) on cloud-free validation pixels.	Partly	IoU: Otsu 0.706, Random Forest 0.658. Both maps were full of speckle (13,479 separate blobs for Otsu). The AI method was behind the threshold.	Clean the maps with morphology.
7	Applied mathematical morphology: opening, closing, removal of small blobs and holes, then the cloud mask again.	Worked	Otsu IoU 0.706 to 0.773 (blobs 13,479 to 54). Random Forest 0.658 to 0.676 (blobs 11,163 to 39). Cleaning helps both, but the forest stays behind.	Look for a better threshold.
8	Chose the threshold at the best point of the ROC curve (0.175 instead of the Otsu value 0.235), then applied a stronger clean-up.	Dropped	IoU 0.855 looked excellent, but the threshold was chosen on the validation pixels themselves, so the score is optimistic.	Redo the evaluation so that nothing is chosen on validation data.
9	Rebuilt everything as one clean notebook: two spatial folds (train on one half, validate on the other, then swap), all metrics on validation pixels only, supervised threshold tuned on the training half. Also fixed the Sentinel-2 reflectance offset, (DN − 1000) / 10000, which the earlier code ignored.	Worked	The Otsu threshold moved from 0.235 to 0.382 (the offset fix is the likely cause). Pooled IoU: Otsu 0.725, tuned threshold 0.700; the original forest scored about 0.71 on validation.	Check whether the Random Forest overfits.
10	Overfitting study: train and validation loss against tree depth and number of trees, then a comparison of five feature sets.	Worked	The original forest (depth 10, leaf 1) had a train-validation IoU gap of 0.063 and 0.076; the regularised one (depth 6, leaf 20) 0.051 and 0.058. Local-context features raised validation IoU from 0.708 to 0.789; NDVI and raw bands did not help.	Apply the same clean-up to every method and compare.
11	Final comparison of the three methods with identical morphology, scored on pooled validation pixels.	Worked	Random Forest 0.873, tuned threshold 0.799, Otsu 0.778 (section 7.1).	Write the report.

The numbers in steps 3 to 8 come from a single validation half and from earlier versions of the code. Only the results in section 7 are comparable across methods.

11.2 Ideas I tried and did not keep

Idea	Why I dropped it	What replaced it
EMS polygons as the target (steps 3 and 4)	Incomplete: no polygons in the middle of the scene	MODIS burned-area polygons
Two-branch cross-training	Random Forest below Otsu, and left and right scores too different to trust	Two spatial folds on the MODIS target

Idea	Why I dropped it	What replaced it
Random Forest on NBR pre, NBR post and dNBR only	Stayed behind Otsu until local-context features were added	Five features including local context
Threshold chosen on the validation pixels	Optimistic score	Threshold tuned on the training half only
Deep forest (depth 10, 1 sample per leaf)	Train-validation IoU gap of 0.063 and 0.076	Depth 6, 20 samples per leaf
Adding NDVI as a feature	No gain (validation IoU 0.712) and a larger gap	Not used
Adding all raw bands (18 features)	Lower IoU than the 5-feature set (0.764 against 0.789) and a gap of 0.077	Not used
Reflectance computed without the L2A offset	Biases the NBR ratio	(DN − 1000) / 10000